

Predicting the Academy Award Winners

Model: Multinomial logit (discrete choice)

The Analytics Edge: Each year the Oscars awards create huge interest in the movie industry, fans and box-office. There is tremendous consequences of taking an Oscar home, be it future earnings or fame. Using a multinomial logit model with data that is available before the awards are given; for example other awards received (Golden Globe), nominations in the Oscars across categories, it is possible to obtain useful predictive models that can predict the winners in four categories - Best Picture, Best Director, Best Leading Actor and Best Leading Actress. This provides an analytics approach to opinions of film critics and expert predictions on Oscar winners.

Oscars (Academy Awards)

The Academy Awards (Oscars) is an annual American award honouring cinematic achievements in films. The awards were first presented in 1929 and is now widely recognised as the most prestigious cinema awards in the world.

Nominations for Oscars:

Currently the Oscar nominations are announced to the public in late January and the awards are presented in late February or early March.

Academy of Motion Picture Arts and Sciences (AMPAS), a professional honorary organization with about 6000 members from different disciplines in film production (such as actors, writers, designers, directors, make-up artists) vote on the nominees and final winners of the Oscar awards.

For most categories, members from each of the branches of AMPAS vote to determine nominees in their categories (for example, directors vote for directors). In special cases such as the Best Picture, all voting members are eligible to select the nominees. From these votes, the top 5 nominees are typically selected to be nominated as Oscar nominees.

The winners are determined by a second round of voting in which all the members are then allowed to vote in most categories.

Viewership of Oscars ceremony

1998 (70th Academy Awards): 57 million viewers
Best Picture: Titanic

2015 (87th Academy Awards): 37 million viewers
Best Picture: Birdman

Impact of Oscars on Movie Performance

www.boxofficemojo.com: Box office performance

In 2016, the weekly earnings for Birdman went up from 1.3 million to 2.5 million in the week following the Oscars. The other big rise in weekly gross for the movie came after it was nominated for the Golden Globe awards.

Similarly in 2014, the Best Picture winner 12 Years A Slave went from 800,000 to 1.5 million to 2.9 million

in the weeks following the oscars.

Winning an Oscar can also help significantly increase the salaries and quality of scripts that actors and actresses receive.

Key Question: Is it possible to predict the winners of Oscars with any reasonable degree of accuracy?

Many people in the media make predictions on the winners of Oscars.

- For example, they might critique the movies, look at the buzz, promotions to make predictions
- Develop data driven models to make predictions

In this, we will consider an approach based on discrete choice models used by I. Pardoe and D. K. Simonton in their paper “Applying discrete choice models to predict Academy Award winners”.

Note that an important aspect of the prediction here is to predict the winner from one of several nominees (typically the winner from 5 nominees).

Multinomial logit model: The IIA property (independence of irrelevant alternatives) is a smaller issue in the case of Oscars predictions since it is not clear if the nominees are close substitutes except if an individual receives multiple nominations in a category in the same year or nominated movies might be from similar genres making them closer substitutes.

Summary

Data: Source is <https://iainpardoe.com/oscars/>. The data consists of details of Oscar nominations and winners including actors, directors and pictures. This includes pre-Oscar awards, number of nominations in the year, cross nominations in other categories, details of previous nominations and if actors won Oscars before and so on.

Model: Multinomial logit is used to predict Oscar winners in terms of these variables.

Decision: The model develops useful predictions that is an alternative to opinions of film critics and media. These predictions have important ramifications on the valuation of people (actors, directors) and the money which the movie makes.

Value: Analytics is crucial in the movie industry. For example, Nate Silver’s website fivethirtyeight.com discusses several models that have been proposed to predict Oscar winners using twitter data, web reviews. Predicting prize winners is important and fun.